MATH 448-548
Midterm project assignment

The best way to submit is to prepare a single .pdf file containing all parts of your project. The file name should be *LASTNAME midterm.pdf*. **Upload your file to Canvas.**

Preparation instructions: write all the codes from scratch. Do not use built-in routines. Your project should contain your code, the output (graphs and tables), and a written part where you do analysis and discuss the results.

Problem 1. Use Taylor theorem to derive the error term for the approximate formula

$$f'(x) \approx \frac{1}{2h}\left(-3f(x) + 4f(x+h) - f(x+2h)\right).$$

given:

$$\boxed{f'(x) \approx \frac{1}{2h}\left[-3f(x) + 4f(x+h) - f(x+2h)\right]} \quad *$$

$$f(x+h)\Big|_{\substack{c=x\\ x-c=h}} = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{6}f'''(x) + k_1 h^4$$

$$f(x+2h)\Big|_{\substack{c=x\\ x-c=2h}} = f(x) + 2hf'(x) + 2h^2 f''(x) + \frac{4}{3}h^3 f'''(x) + k_2 h^4$$

$$\frac{1}{2h}\Big[-3\cancel{f(x)} + \cancel{4f(x)} + 4hf'(x) + 2h^2\cancel{f''(x)} + \frac{4h^3}{6}f'''(x) + 4k_1 h^4$$

$$- \cancel{f(x)} - 2hf'(x) - 2h^2\cancel{f''(x)} - \frac{4}{3}h^3 f'''(x) - k_2 h^4\Big)$$

$$f'(x) - \frac{1}{3}h^2 + O(h^3)$$

$$\text{error} \approx O(h^3)$$

Problem 2. Consider the following variation of the Newton's method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)}.$$

Find constants C and s such that

$$e_{n+1} = Ce_n^s.$$

Problem 3. Consider the linear system $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 6.25 & -1 & 0.5 \\ -1 & 5 & 2.12 \\ 0.5 & 2.12 & 3.6 \end{bmatrix},$$

and

$$\mathbf{b} = \begin{bmatrix} 7.5 \\ -8.68 \\ -0.24 \end{bmatrix}.$$

Write a computer program for LU-factorization with a unit lower triangular L (meaning that the diagonal entries should be equal to one). Then write a program for the Cholesky factorization. **WARNING: avoid using shortcuts. The programming should be done "from scratch"**.

$$\begin{bmatrix} 1 & \cdot & \cdot \\ l_{21} & 1 & \cdot \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ \cdot & u_{22} & u_{23} \\ \cdot & \cdot & u_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$(1,1)$ $u_{11} = a_{11}$

$(1,2)$ $u_{12} = a_{12}$

$(1,3)$ $u_{13} = a_{13}$

$(1,:)$ I

$u[1,j] = a(1,j)$, $j = 1:N$

$u[2,j] = a[2,j] - \sum_{k=1}^{k=1} l[2,k] \cdot u[k,j]$, $j = 2:N$

$(2,1)$ $l_{21} u_{11} = a_{21} \Rightarrow l_{21} = \dfrac{a_{21}}{u_{11}}$

$(2,2)$ $l_{21} u_{12} + u_{22} = a_{22}$

$u_{22} = a_{22} - l_{21} u_{12}$

$(2,3)$ $l_{21} \cdot u_{13} + u_{23} = a_{23}$

$u_{23} = a_{23} - l_{21} \cdot u_{13}$

Problem 4. Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - |x - \frac{1}{2}|.\right)$$

Do the following.

1. Rewrite the equation as a fixed point problem for a function $T(x)$ related to $f(x)$. Prove that $T(x)$ (i) maps the interval $[0.45, 0.55]$ into itself, and (ii) $T(x)$ is contractive on this interval. Based on the theorems in the books and notes, make conclusions about existence and uniqueness of a solution to $f(x) = 0$.

Problem 4. Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - |x - \frac{1}{2}|.\right)$$

Do the following.

2. Write the computer program implementing the fixed point iteration $x_{n+1} = T(x_n)$. Discuss the choice of an initial guess x_0. Calculate the solution to within 10 decimal places. Note the number of iterations needed.

Problem 4. Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - |x - \frac{1}{2}|.\right)$$

Do the following.

3. Use a priori and a posteriori error estimates from the notes to estimate the number of iterations needed to obtain the above accuracy. Which estimate is more accurate?

Problem 4. Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - |x - \frac{1}{2}|.\right)$$

Do the following.

4. Implement the Newton's method on a computer for the same equation starting with x_0 within the interval $[0.45, 0.55]$. Discuss the theoretical difficulty related to non-existence of $f'(x)$ for certain point(s) within the interval. How would you set up the Newton's method to resolve the problem?

Problem 4. Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - |x - \frac{1}{2}|.\right)$$

Do the following.

5. Note the number of iterations needed to calculate the solution to within 10 decimal places using the Newton's method. Discuss the speed of convergence of the Newton's method and compare it with the convergence speed of the fixed point iterations.

Problem 4. Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - |x - \frac{1}{2}|.\right)$$

Do the following.

6. Implement the alternative error control strategy proposed by you in HW 3, Problem 2. Compare the results with the ones obtained using a priori and a posteriori error estimates.

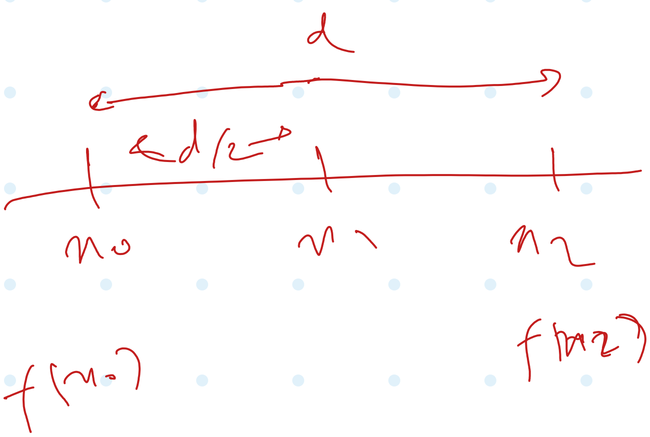
Problem 5. Implement and analyze the Ridders' method for solving the equation $f(x) = 0$. For guidance, you might wish to read the Wikipedia page on the Ridders' method. Also feel free to consult other sources.

Brief Description of the method. Given the initial bracketing interval $[x_0, x_2]$, containing **one** solution, and such that $f(x_0)$ and $f(x_2)$ have opposite signs, compute the midpoint $x_1 = (x_0 + x_2)/2$. Then define a new function $h(x) = f(x)e^{ax}$. Find the parameter a that ensures $h(x_1) = \frac{1}{2}(h(x_0) + h(x_2))$. Then define x_3 as the x-intercept of the line passing through $(x_0, h(x_0))$ and $(x_2, h(x_2))$. Use x_3 as one of the endpoints of the next interval bracketing the solution. The other endpoint is x_1 if $f(x_1)f(x_3) < 0$. Otherwise, choose either x_0 or x_2 based on the requirement that the sign of $f(x)$ at the chosen point must be opposite to the sign of $f(x_3)$. Continue until the desired accuracy is reached.

Questions and items to implement

1. Show that a is uniquely defined and give the equation for finding it.

$\exists\ x^{*}$ s.t. $f(x^*) = 0$.



$x_1 = \frac{x_0 + x_2}{2}$

$$f(x_1)\cdot e^{ax_1} = \frac{1}{2}\left(f(x_0)e^{ax_0} + f(x_2)\cdot e^{ax_2}\right)$$

a

$$f_1 e^{ax_1} = \frac{f_0}{2} e^{a(x_1 - \frac{d}{2})} + \frac{f_2}{2} e^{a(x_1 + \frac{d}{2})}$$

$\div\ e^{ax_1}$:

$$f_1 = \frac{f_0}{2} e^{-\frac{ad}{2}} + \frac{f_2}{2} e^{\frac{ad}{2}}$$

$$\times\ e^{\frac{ad}{2}}$$

$$f_1 e^{\frac{ad}{2}} = \frac{f_0}{2} + \frac{f_2}{2} e^{ad}$$

$$\left(\frac{f_2}{2}\right) e^{ad} - f_1 e^{\frac{ad}{2}} + \frac{f_0}{2} = 0$$

$$e^{\frac{ad}{2}} = \frac{f_1 \pm \overbrace{\sqrt{(\underbrace{f_1^2}_{+} - \underbrace{f_0 f_2}_{+})}}^{(f_1 + sm)}}{f_2}$$

if $f_2 > 0$:

$$(1a)\quad e^{\frac{ad}{2}} = \frac{f_1 + \sqrt{f_1^2 - f_0 f_2}}{f_2}$$

else if $f_2 < 0$:

1b $\quad e^{\frac{ad}{2}} = \dfrac{f_1 - \sqrt{f_1^2 - f_0 f_2}}{f_2}$

1a $\Rightarrow$ $\quad a = \dfrac{2}{d} \times \ln\left(\dfrac{f_1 + \sqrt{f_1^2 - f_0 f_2}}{f_2}\right)$

$f_2 > 0$

1b $\Rightarrow$ $\quad a = \dfrac{2}{d} \ln\left(\dfrac{f_1 - \sqrt{f_1^2 - f_0 f_2}}{f_2}\right)$

$f_2 < 0$

or $$\boxed{a = \frac{2}{d} \ln\left(\frac{f_1 + \operatorname{sgn}(f_2)\sqrt{f_1^2 - f_0 f_2}}{f_2}\right)}$$

or alternatively,

or $$\boxed{a = \frac{2}{d} \ln\left(\frac{f_1 - \operatorname{sgn}(f_0)\sqrt{f_1^2 - f_0 f_2}}{f_2}\right)}$$

$\because f_0 \cdot f_2 < 0$

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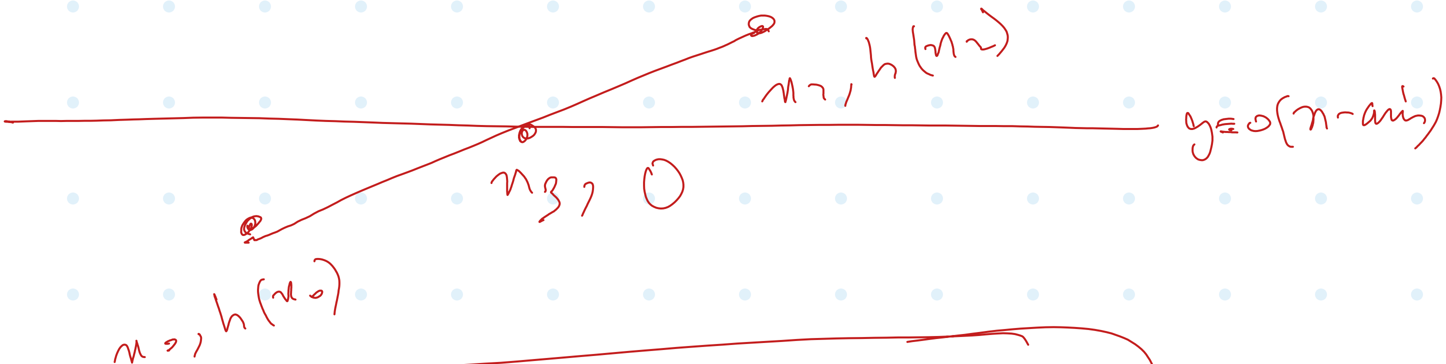
2. Work out a formula for x_3 in detail.



$$\boxed{m = \frac{h(x_2) - h(x_1)}{x_2 - x_1}}$$

$$h(x_1) = m x_1 + c$$

$$\boxed{m = \frac{h_2 - h_1}{d/2}}$$

$$h(x_1) - m x_1 = c$$

$$\boxed{h_1 - \left(\frac{h_2 - h_1}{d/2}\right) x_1 = c}$$

$$\boxed{\sigma \quad \mu_1\left(1+\frac{x_1}{d/2}\right) - \left(\frac{x_1}{d/2}\right)\mu_2 = c} \quad \triangle$$

$$0 = m\,x_3 + c$$

$$x_3 = -\frac{c}{m}$$

$$x_3 = \frac{\left(\frac{x_1}{d/2}\right)\mu_2 - \left(1+\frac{x_1}{d/2}\right)\mu_1}{\frac{\mu_2-\mu_1}{d/2}}$$

$$\sigma \quad x_3 = \frac{x_1\mu_2 - \left(\frac{d}{2}+x_1\right)\mu_1}{\mu_2-\mu_1}$$

$$\boxed{\sigma \quad x_3 = \frac{x_1\mu_2 - x_2\mu_1}{\mu_2-\mu_1}} \quad *$$

$$\Rightarrow x_3 = \frac{x_1 M_2 - x_1 M_1}{M_2 - M_1} - \frac{\frac{d}{2} \cdot M_1}{M_2 - M_1}$$

$$\Rightarrow x_3 = x_1 - \frac{d}{2} \frac{M_1}{M_2 - M_1}$$

$$\Rightarrow x_3 = x_1 - \frac{d}{2} \frac{1}{\frac{M_2}{M_1} - 1}$$

$$\Rightarrow \boxed{x_3 = x_1 + \frac{d}{2} \frac{1}{1 - M_2/M_1}}$$

Problem 5. Implement and analyze the Ridders' method for solving the equation $f(x) = 0$. For guidance, you might wish to read the Wikipedia page on the Ridders' method. Also feel free to consult other sources.

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3. Implement Ridders' method on a compuater together with the standard bisection method for finding **all** solutions of
$$e^x - x^2 = 0.$$

Problem 5. Implement and analyze the Ridders' method for solving the equation $f(x) = 0$. For guidance, you might wish to read the Wikipedia page on the Ridders' method. Also feel free to consult other sources.

Brief Description of the method. Given the initial bracketing interval $[x_0, x_2]$, containing **one** solution, and such that $f(x_0)$ and $f(x_2)$ have opposite signs, compute the midpoint $x_1 = (x_0 + x_2)/2$. Then define a new function $h(x) = f(x)e^{ax}$. Find the parameter a that ensures $h(x_1) = \frac{1}{2}(h(x_0) + h(x_2))$. Then define x_3 as the x-intercept of the line passing through $(x_0, h(x_0))$ and $(x_2, h(x_2))$. Use x_3 as one of the endpoints of the next interval bracketing the solution. The other endpoint is x_1 if $f(x_1)f(x_3) < 0$. Otherwise, choose either x_0 or x_2 based on the requirement that the sign of $f(x)$ at the chosen point must be opposite to the sign of $f(x_3)$. Continue until the desired accuracy is reached.

4. Numerically, compare the convergence rate of both algorithms. Discuss the results. What is the approximate order of convergence for the Ridders' algorithm?